

Documentation: Command Line Editor REPL System

Introduction

This project implements a console-based command-line editor in C++ with enhanced input handling, including:

- Arrow key navigation
- Special key commands (F1-F10)
- Cursor-aware input editing
- A REPL that supports interactive editing.

The system mimics retro-style BASIC terminal interfaces and is suitable for embedding into interpreters or command shells.

Dependencies

Headers & Libraries:

- `<iostream>`: Standard console I/O
- `<fstream>`: File I/O (currently unused)
- `<string>`: String handling
- `<conio.h>`: Low-level input for Windows
- `SystemInterface.h`: Console I/O abstraction
- `CommandLineEditor.h`: Input editing with cursor
- `ScreenRenderer.h`: Redraws console buffer
- `SpecialKeyHandler.h`: Maps F1-F10 to commands

Class Descriptions

- `CommandLineEditor`: Handles user input, editing, and cursor navigation
- `ScreenRenderer`: Manages output and redraws
- `SpecialKeyHandler`: Maps function keys to commands
- `SystemInterface`: Provides platform-specific I/O abstraction

Public Methods

CommandLineEditor:

- `getLineFromCli(std::string&)`: Reads user input interactively
- `setSpecialKeyHandler(...)`: Attaches special key handler

ScreenRenderer:

- `moveCursor(int)`: Moves cursor within input
- `redrawBuffer(...)`: Redraws current buffer with cursor

****SpecialKeyHandler**:**

- handleSpecialKey(...): Handles function keys and injects commands

****SystemInterface**:**

- init(): Setup stub
- readKey(): Reads a character or key
- putChar(): Prints one char
- getChar(): Reads one char using `_getch()`
- printString(): Prints full string
- clearScreen(): ANSI clear
- moveCursor(x, y): Positions cursor

Internal Data Members

****CommandLineEditor**:**

- buffer: Current input
- cursorPosition: Index in buffer
- specialKeyHandler: Handles F1-F10

****check_SKey (Struct)**:**

- isSpecial: True if key is special
- ch: Normal character
- sKey: Enum for F1-F10, arrows

Notes

- ANSI cursor movement
- Uses `_getch()`, so Windows-only
- ScreenRenderer keeps CommandLineEditor simple
- Designed for BASIC-style REPLs or interpreters